03

MGU BCA - 3RD SEM

Quantitative Techniques

Module 3 - Correlation and Regression

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Complete question bank: every important, repeated and sure-mark question for this module, each with a full-mark model answer. Nothing important left out.

EXAM PATTERN (from real MGU question papers)

EXAM PATTERN (QT): The real MG3CCRBCA200 paper pattern is UNKNOWN - no Honours model paper and no past BCA paper exists online (first exam 22-10-2025 was never published). Closest evidence: MGU B.Com 'Quantitative Techniques for Business' papers (same university, overlapping statistics syllabus) run 80 marks / 3 hours: Part A answer any 10 x 2 marks = 20 (short theory + one-line numericals), Part B answer any 6 x 5 marks = 30 (medium theory + numericals), Part C answer any 2 x 15 marks = 30 (long numericals: combined mean/SD/CV two-firm problem repeated 3x, Bayes theorem). QT is a 4-credit course, so the Honours paper may instead follow a 70-mark / 2-hour format (like DBMS) - this is NOT confirmed. Prepare every module: Part A 2-mark questions span all modules.

TOPIC 1

Correlation: Definition and Types

1

Correlation measures how strongly two variables move together.

Positive: both rise together, like height and weight. Negative: one rises while the other falls, like price and demand. Zero: no relation, like shoe size and marks.

Linear correlation follows a straight-line pattern; non-linear follows a curve.

QUESTION BANK

1 **Define correlation and explain its types with examples.**

IMPORTANT

Correlation is a statistical measure that shows how strongly two variables move together.

It tells us whether a change in one variable is linked with a change in the other.

Positive correlation: both variables increase together. Example: height and weight of students.

Negative correlation: one variable increases while the other decreases. Example: price of a product and its demand.

Zero correlation: there is no relationship between the two variables. Example: shoe size and exam marks.

The value of correlation lies between -1 and +1.

Correlation is a very useful tool in business, economics and science to find hidden relationships.

2 Distinguish between linear and non-linear correlation.

IMPORTANT

Linear correlation exists when the two variables change in a constant ratio.

In linear correlation the points on a scatter diagram lie close to a straight line.

Example: if income rises by Rs. 1000 every year and savings rise by Rs. 200 every year, the relation is linear.

Non-linear correlation exists when the ratio of change between the two variables is not constant.

In non-linear correlation the points on a scatter diagram form a curve, not a straight line.

Example: at first a small rise in advertising gives a big rise in sales, but later extra advertising gives very little extra sales.

So the key difference is a straight-line pattern versus a curved pattern.

3 Explain the meaning and significance of the concept of correlation. Does it always imply cause and effect? (2024)

IMPORTANT

Correlation measures the strength and direction of the linear relationship between two variables, with r lying between -1 and +1.

It is significant because it lets us predict one variable from another, like estimating sales from advertising spend.

It helps managers, scientists and economists discover hidden relationships in data before taking decisions.

However, correlation does NOT always imply cause and effect.

Two variables can move together because of a third common cause: ice-cream sales and drowning deaths both rise in summer, but one does not cause the other.

Such misleading cases are called spurious or nonsense correlation.

Therefore a high correlation is a clue, not a proof; causation needs controlled study.

In short, correlation shows association between variables, but cause and effect must be established separately.

TOPIC 2

2

Scatter Diagram

A scatter diagram plots each (x, y) pair as a dot on a graph.

Dots rising left to right show positive correlation; falling dots show negative correlation.

Tightly packed dots mean strong correlation; widely spread dots mean weak or no correlation.

QUESTION BANK

4 **What is a scatter diagram? How do you read correlation from it?**

IMPORTANT

A scatter diagram is a graph on which each pair of (x, y) values is plotted as a dot.

It is the simplest visual method to see the relationship between two variables.

To read correlation from it, look at the direction and closeness of the dots.

Dots rising from left to right show positive correlation.

Dots falling from left to right show negative correlation.

Dots scattered all over with no pattern show zero or no correlation.

Tightly packed dots in a narrow band mean strong correlation; widely spread dots mean weak correlation.

5 **What are the merits and limitations of the scatter diagram?**

IMPORTANT

Merits:

A scatter diagram is very simple to draw and easy to understand.

It shows the pattern and direction of the relationship at a glance.

Odd or unusual values stand out clearly on the diagram.

It needs no complicated calculation.

Limitations:

It gives only a rough idea and no exact numerical measure of correlation.

Two people may judge the same diagram differently, so it is not fully objective.

It is not useful when the number of observations is very large.

Despite its limits, it is the best first step before calculating the correlation coefficient.

TOPIC 3

3

Karl Pearson's Coefficient of Correlation

Karl Pearson's r = [N*sum(xy) - sum(x)*sum(y)] / sqrt([N*sum(x^2) - (sum(x))^2] * [N*sum(y^2) - (sum(y))^2]).

r always lies between -1 and +1. +1 is perfect positive, -1 is perfect negative, 0 is no linear relation.

It needs both variables in numbers and is affected by extreme values.

QUESTION BANK

6 **Calculate the coefficient of correlation for x: 1,2,3,4,5 and y: 2,3,5,4,6. (2024, 2022)**

SURE MARK

Formula: r = [N*sum(xy) - sum(x)*sum(y)] / sqrt([N*sum(x^2) - (sum(x))^2] * [N*sum(y^2) - (sum(y))^2])

Here N = 5.

sum(x) = 1+2+3+4+5 = 15.

sum(y) = 2+3+5+4+6 = 20.

sum(xy) = (1*2)+(2*3)+(3*5)+(4*4)+(5*6) = 2+6+15+16+30 = 69.

sum(x^2) = 1+4+9+16+25 = 55.

sum(y^2) = 4+9+25+16+36 = 90.

Numerator = N*sum(xy) - sum(x)*sum(y) = 5*69 - 15*20 = 345 - 300 = 45.

First term of denominator = N*sum(x^2) - (sum(x))^2 = 5*55 - 225 = 275 - 225 = 50.

Second term of denominator = N*sum(y^2) - (sum(y))^2 = 5*90 - 400 = 450 - 400 = 50.

Denominator = sqrt(50 * 50) = 50.

r = 45 / 50 = 0.9.

Answer: r = 0.9, which means strong positive correlation.

7 State the formula of Karl Pearson's coefficient of correlation and its properties.

IMPORTANT

Karl Pearson's coefficient of correlation is denoted by r and its formula is:

r = [N*sum(xy) - sum(x)*sum(y)] / sqrt([N*sum(x^2) - (sum(x))^2] * [N*sum(y^2) - (sum(y))^2])

Properties:

r always lies between -1 and +1.

r = +1 means perfect positive correlation and r = -1 means perfect negative correlation.

r = 0 means there is no linear relationship between the variables.

r measures only linear relation; it cannot catch a curved relationship.

It has no unit, because it is a pure number.

It needs both variables to be in numerical form.

It is affected by extreme values, so one odd value can change r a lot.

8 Explain the underlying assumptions of Karl Pearson's co-efficient of correlation. (2024)

IMPORTANT

Karl Pearson's coefficient r is valid only when certain assumptions about the data hold.

First, the relationship between the two variables must be linear; r cannot measure a curved relationship properly.

Second, both variables must be measured on an interval or ratio scale, that is, proper numerical data.

Third, the observations must be meaningfully paired, with a sensible link between each x and its y.

Fourth, the data should be free from extreme values, because a single outlier can distort r heavily.

Fifth, for full reliability the variables are assumed to follow roughly a normal distribution.

If these assumptions fail, rank correlation or other non-parametric methods are safer.

In short, Pearson's r assumes linear, numerical, outlier-free paired data.

9 **If the covariance between X and Y is 488 and variance of X and Y are 824 and 325 respectively, find out coefficient of correlation. (2022)**

IMPORTANT

The shortcut formula is r = Cov(X,Y) / sqrt(VarX x VarY).

Here Cov = 488, VarX = 824 and VarY = 325.

Denominator = sqrt(824 x 325) = sqrt(267800) = 517.50.

r = 488 / 517.50 = 0.94.

The value is close to +1, showing a high degree of positive correlation between X and Y.

Answer: the coefficient of correlation is 0.94.

10 **Compute Pearson's coefficient of correlation from the following data taking 60 and 70 as the assumed averages of X and Y respectively. Also see if it comes between 0.85 and 0.95. X: 45, 55, 56, 58, 60, 65, 68, 70, 75, 80, 85; Y: 56, 50, 48, 60, 62, 64, 65, 70, 74, 82, 90. (2022)**

IMPORTANT

With assumed means Ax = 60 and Ay = 70, take dx = X - 60 and dy = Y - 70 for the 11 pairs.

Sums: sum dx = 57, sum dy = -49, sum dx-square = 1709, sum dy-square = 1865, sum dx.dy = 1148.

r = [n.sum dx.dy - sum dx.sum dy] / sqrt[(n.sum dx-square - (sum dx)^2)(n.sum dy-square - (sum dy)^2)].

Numerator = 11x1148 - 57x(-49) = 12628 + 2793 = 15421.

Denominator = sqrt[(18799 - 3249)(20515 - 2401)] = sqrt(15550 x 18114) = 16783.2.

r = 15421 / 16783.2 = 0.92.

0.92 lies between 0.85 and 0.95, so the check is satisfied.

Answer: r = 0.92, which is between 0.85 and 0.95.

TOPIC 4

4

Degree of Correlation, Probable Error and Coefficient of Determination

Degree of correlation: |r| near 1 is high, near 0.5 is moderate, near 0 is low or none.

Probable error: PE = 0.6745 * (1 - r^2) / sqrt(N). If r > 6*PE, the correlation is significant.

Coefficient of determination = r^2. It tells what fraction of the variation in y is explained by x.

QUESTION BANK

11 What is probable error? How do you interpret correlation using it?

IMPORTANT

Probable error (PE) is a measure used to test whether an observed correlation is real or just accidental.

Its formula is: PE = 0.6745 * (1 - r^2) / sqrt(N).

Here r is the correlation coefficient and N is the number of pairs.

Interpretation rules:

If r is greater than 6 times PE, the correlation is significant and definitely exists.

If r is less than PE, there is practically no correlation.

If r lies between PE and 6 times PE, we cannot say anything surely; the data may be small.

Probable error is useful because a high r from a small sample can still be doubtful.

12 If r = 0.9 and N = 25, find the probable error and comment.

IMPORTANT

Formula: PE = 0.6745 * (1 - r^2) / sqrt(N).

Given r = 0.9 and N = 25.

r^2 = 0.9 * 0.9 = 0.81.

1 - r^2 = 1 - 0.81 = 0.19.

sqrt(N) = sqrt(25) = 5.

PE = 0.6745 * 0.19 / 5.

First multiply: 0.6745 * 0.19 = 0.128155.

Then divide: 0.128155 / 5 = 0.025631.

So PE = 0.0256 approx.

6 * PE = 6 * 0.0256 = 0.1536.

Since r = 0.9 is greater than 6*PE = 0.1536, the correlation is highly significant.

Answer: PE = 0.0256; the correlation is significant.

13 **What is the coefficient of determination? If r = 0.8, interpret it.**

IMPORTANT

The coefficient of determination is the square of the correlation coefficient, r^2.

It tells what fraction of the variation in one variable is explained by the other variable.

Its value lies between 0 and 1, and it is often written as a percentage.

For r = 0.8:

r^2 = 0.8 * 0.8 = 0.64.

This means 64 percent of the variation in y is explained by the variation in x.

The remaining 36 percent of the variation is due to other factors not included in the model.

So a higher r^2 means a better and more useful relationship for prediction.

TOPIC 5

Spearman's Rank Correlation and Tie in Ranks

5

Spearman's rank correlation is used when data is in ranks (1st, 2nd...) or not normally distributed.

rho = 1 - [6 * sum(d^2)] / [n * (n^2 - 1)], where d is the difference of the two ranks.

Tie in ranks: when values repeat, give the average rank and add (t^3 - t)/12 to sum(d^2) for each tie of size t.

QUESTION BANK

14 **State Spearman's rank correlation formula. When is it used? (2024, 2022)**

REPEATED

Spearman's rank correlation coefficient is denoted by rho and its formula is:

rho = 1 - [6 * sum(d^2)] / [n * (n^2 - 1)]

Here n is the number of pairs and d is the difference between the two ranks of each item.

When it is used:

It is used when data is given in the form of ranks, like 1st, 2nd, 3rd.

It is used for qualitative data that can be ranked, like beauty, honesty or taste.

It is also used when the data is not normally distributed or has extreme values.

Example: to find the relationship between the ranks of students in Maths and in Statistics.

15 **Ranks in Maths: 1,2,3,4,5; ranks in Stats: 2,1,4,3,5. Find the rank correlation.**

IMPORTANT

Formula: rho = 1 - [6 * sum(d^2)] / [n * (n^2 - 1)]

Here n = 5.

Find d = Maths rank minus Stats rank for each student:

Student 1: d = 1 - 2 = -1, d^2 = 1.

Student 2: d = 2 - 1 = 1, d^2 = 1.

Student 3: d = 3 - 4 = -1, d^2 = 1.

Student 4: d = 4 - 3 = 1, d^2 = 1.

Student 5: d = 5 - 5 = 0, d^2 = 0.

sum(d^2) = 1 + 1 + 1 + 1 + 0 = 4.

Numerator = 6 * 4 = 24.

Denominator = 5 * (25 - 1) = 5 * 24 = 120.

rho = 1 - 24/120 = 1 - 0.2 = 0.8.

Answer: rho = 0.8, which means strong positive rank correlation.

16 **What is a tie in ranks? How is it corrected?**

IMPORTANT

A tie in ranks happens when two or more items have the same value, so they share the same rank.

Example: if two students tie for the 2nd and 3rd places, each is given the average rank 2.5.

The average rank given to the tied items keeps the total of all ranks correct.

Correction method:

For every group of ties, add (t^3 - t)/12 to sum(d^2).

Here t is the number of tied items in that group.

If there are two tie groups, apply this correction for each group separately.

The corrected sum(d^2) is then used in the formula rho = 1 - [6 * sum(d^2)] / [n * (n^2 - 1)].

17 **Write down any three merits and demerits of Spearman's rank correlation co-efficient. (2024)**

IMPORTANT

Merits: first, it is simple to understand and easy to calculate, simpler than Karl Pearson's method.

Second, it can be used even when the data is qualitative, like ranks given for beauty, honesty or intelligence.

Third, it is not affected much by extreme values, since only ranks matter, not actual magnitudes.

Demerits: first, it cannot be applied to grouped frequency distributions.

Second, it is less precise and less powerful than Pearson's coefficient for numerical data.

Third, when the number of observations is large, ranking all of them becomes tedious.

In short, rank correlation is simple and flexible, but it is less exact than Pearson's method.

TOPIC 6

6

Regression: Definition, Types and Regression Lines

Regression studies how one variable depends on another; it is used for prediction.

Types: linear vs non-linear; simple (one independent variable) vs multiple.

Regression lines: y on x predicts y from x; x on y predicts x from y. They always cross at (xbar, ybar).

QUESTION BANK

18 **Define regression. Explain its types.**

IMPORTANT

Regression is a statistical method that studies the average relationship between variables for estimation and prediction.

While correlation only tells the strength of a relationship, regression gives an actual equation to predict one variable from another.

Types of regression:

Linear regression: the relationship is a straight line. Example: sales rise steadily with advertising.

Non-linear regression: the relationship is a curve. Example: production rises fast at first and then slows down.

Simple regression: there is one independent variable. Example: predicting marks (y) from study

hours (x).

Multiple regression: there are two or more independent variables. Example: predicting marks (y) from study hours (x1) and attendance (x2).

Example: a shop uses regression to predict next month's sales (y) from this month's advertising spend (x).

19 **What are regression lines? What is special about their intersection?**

IMPORTANT

There are two regression lines:

The line of regression of y on x, which is used to predict y from a given x.

The line of regression of x on y, which is used to predict x from a given y.

Each line is the best fitting line for its own direction of prediction.

Special point about their intersection:

The two regression lines always cross each other at the point (xbar, ybar).

This point is the average of x and the average of y taken together.

If r = +1 or r = -1, that is perfect correlation, the two lines merge into a single line.

The smaller the angle between the two lines, the higher the correlation.

TOPIC 7

Regression Equations of y on x and x on y

7

y on x: y - ybar = byx * (x - xbar), where byx = r * (SDy / SDx).

x on y: x - xbar = bxy * (y - ybar), where bxy = r * (SDx / SDy).

byx and bxy are the regression coefficients (the slopes of the lines).

20 For x: 1,2,3,4,5 and y: 2,3,5,4,6, find the regression line of y on x. (2022)

REPEATED

Formula for the regression line of y on x: y - ybar = byx * (x - xbar).

First find the means:

xbar = (1+2+3+4+5)/5 = 15/5 = 3.

ybar = (2+3+5+4+6)/5 = 20/5 = 4.

Now find byx = sum((x - xbar)(y - ybar)) / sum((x - xbar)^2).

Numerator terms (x - xbar)(y - ybar):

(1-3)(2-4) = (-2)(-2) = 4.

(2-3)(3-4) = (-1)(-1) = 1.

(3-3)(5-4) = 0*1 = 0.

(4-3)(4-4) = 1*0 = 0.

(5-3)(6-4) = 2*2 = 4.

Numerator = 4 + 1 + 0 + 0 + 4 = 9.

Denominator terms (x - xbar)^2:

(1-3)^2 = 4, (2-3)^2 = 1, (3-3)^2 = 0, (4-3)^2 = 1, (5-3)^2 = 4.

Denominator = 4 + 1 + 0 + 1 + 4 = 10.

byx = 9/10 = 0.9.

Line: y - 4 = 0.9 * (x - 3).

y = 0.9x - 2.7 + 4.

y = 0.9x + 1.3.

Answer: the regression line of y on x is y = 0.9x + 1.3.

21 Write the two regression equations and define the regression coefficients.

IMPORTANT

The two regression equations are:

Regression of y on x: y - ybar = byx * (x - xbar).

Regression of x on y: x - xbar = bxy * (y - ybar).

Here xbar and ybar are the means of x and y.

Regression coefficients:

byx is the regression coefficient of y on x. Its formula is byx = r * (SDy / SDx).

It is the slope of the line of y on x and is used when predicting y.

bxy is the regression coefficient of x on y. Its formula is bxy = r * (SDx / SDy).

It is the slope of the line of x on y and is used when predicting x.

Both coefficients always have the same sign as the correlation coefficient r.

22 Write a note on least square method of computing regression equation. (2022)

IMPORTANT

The least square method finds the regression line that fits the observed data best.

'Best' here means the line that makes the sum of the squared vertical distances of the points from the line as small as possible.

For y on x, we choose y = a + bx so that sum(y - a - bx)^2 is minimum.

Minimising gives the two normal equations: sum y = n.a + b.sum x and sum xy = a.sum x + b.sum x-square.

Solving these two equations gives the constants a and b, and hence the regression line.

The fitted line always passes through the point (mean of x, mean of y).

It is the most scientific method of fitting a regression line.

In short, least squares picks the line with the minimum total squared error.

23 What are the features of regression coefficients? (2022)

IMPORTANT

The two regression coefficients byx and bxy have the following features.

First, both coefficients always carry the same sign, which is the sign of the correlation coefficient r.

Second, r is the geometric mean of the two coefficients: r = plus or minus sqrt(byx x bxy).

Third, if one regression coefficient is greater than 1, the other must be less than 1, because their product r-square cannot exceed 1.

Fourth, the coefficients are not symmetric: byx is generally not equal to bxy.

Fifth, both regression lines intersect at the point (x-bar, y-bar), the means of the two variables.

In short, the regression coefficients share the sign of r and are linked through r-square.

24 **Given the following data, what would be the possible yield of rice per acre when rainfall is 29 cm? Coefficient of correlation between rainfall and yield is 0.8. Rainfall: Mean 25, Variance 9; Yield: Mean 40, Variance 36. (2022)**

IMPORTANT

Let x = rainfall and y = yield. SDx = sqrt(9) = 3 and SDy = sqrt(36) = 6.

Regression coefficient of y on x: byx = r x (SDy/SDx) = 0.8 x (6/3) = 1.6.

Since rainfall is the independent variable, use the regression line of y on x.

Regression line of y on x: y - y-bar = byx(x - x-bar), so y - 40 = 1.6(x - 25).

Put x = 29: y - 40 = 1.6 x 4 = 6.4, giving y = 46.4.

This estimate is reliable because the correlation 0.8 is fairly high.

Answer: the expected yield of rice is 46.4 per acre when rainfall is 29 cm.

25 **Following data relates to marks in Accounts and Statistics: Accounts - Mean 30, SD 10; Statistics - Mean 35, SD 7; Coefficient of correlation 0.8. Find two regression equations and calculate the expected marks in Accounts if marks secured by a student in Statistics are 40. (2022)**

IMPORTANT

Let x = Accounts marks (mean 30, SD 10) and y = Statistics marks (mean 35, SD 7), with r = 0.8.

byx = r x (SDy/SDx) = 0.8 x (7/10) = 0.56; regression of y on x: y - 35 = 0.56(x - 30).

bxy = r x (SDx/SDy) = 0.8 x (10/7) = 1.143; regression of x on y: x - 30 = 1.143(y - 35).

To estimate Accounts (x) from Statistics (y = 40), use the x-on-y line.

x - 30 = 1.143 x (40 - 35) = 1.143 x 5 = 5.71, so x = 35.71.

Answer: the two regression equations are y - 35 = 0.56(x - 30) and x - 30 = 1.143(y - 35); expected Accounts marks = 35.71.

26 **A panel of judges A and B graded seven debaters and independently awarded the following marks: A: 40, 34, 28, 30, 44, 38, 31; B: 32, 39, 26, 30, 38, 34, 28. An eighth debater was awarded 36 marks by Judge A while Judge B was not present. How many marks would you expect Judge B to award? (2022)**

IMPORTANT

This is a regression estimate: predict B's marks (y) from A's marks (x) using the y-on-x line.

Sums for n = 7: sum x = 245, sum y = 227, sum x-square = 8781, sum xy = 8066.

byx = [n.sum xy - sum x.sum y] / [n.sum x-square - (sum x)^2] = [56462 - 55615] / [61467 - 60025] = 847/1442 = 0.587.

Means: x-bar = 245/7 = 35, y-bar = 227/7 = 32.43.

Regression line: y - 32.43 = 0.587(x - 35).

For x = 36: y = 32.43 + 0.587 x 1 = 33.02.

Answer: Judge B would be expected to award about 33 marks.

TOPIC 8

8

Relation between Correlation and Regression Coefficients

r^2 = byx * bxy, so r = +sqrt(byx*bxy) or -sqrt(byx*bxy), taking the sign of the coefficients.

Both regression coefficients always have the same sign as r.

To identify the lines: the coefficient of x in the "y on x" equation is byx; the coefficient of y in the "x on y" equation is bxy.

QUESTION BANK

27 **Show that r^2 = byx * bxy.**

IMPORTANT

The two regression coefficients are defined as:

byx = r * (SDy / SDx).

bxy = r * (SDx / SDy).

Multiply the two coefficients:

byx * bxy = [r * (SDy / SDx)] * [r * (SDx / SDy)].

Group the terms: = r * r * (SDy / SDx) * (SDx / SDy).

The SD terms cancel: (SDy / SDx) * (SDx / SDy) = 1.

So byx * bxy = r^2.

This proves that the product of the two regression coefficients equals the square of the correlation coefficient.

Taking the square root: r = +sqrt(byx * bxy) or -sqrt(byx * bxy), with the sign of the coefficients.

28 **If byx = 0.9 and bxy = 0.9, find r.**

IMPORTANT

Formula: r^2 = byx * bxy.

Substitute the values: r^2 = 0.9 * 0.9 = 0.81.

Take the square root: r = +sqrt(0.81) or -sqrt(0.81).

sqrt(0.81) = 0.9, so r = +0.9 or -0.9.

Both regression coefficients are positive, so r must also be positive.

Answer: r = 0.9.

29 **Given 2x - y = 5 and 3x - 2y = 4, identify the regression lines and find r.**

IMPORTANT

We must decide which line is y on x and which is x on y, because r^2 = byx * bxy must be between 0 and 1.

First try: treat 2x - y = 5 as the line of y on x.

Rewrite as y = 2x - 5, so byx = 2.

Treat 3x - 2y = 4 as the line of x on y: 3x = 2y + 4, so x = (2/3)y + 4/3, giving bxy = 2/3.

Then r^2 = 2 * (2/3) = 4/3 = 1.33.

This is greater than 1, which is impossible, so this assignment is wrong.

So swap the roles: treat 2x - y = 5 as the line of x on y.

2x = y + 5, so x = 0.5y + 2.5, giving bxy = 0.5.

Treat 3x - 2y = 4 as the line of y on x.

2y = 3x - 4, so y = 1.5x - 2, giving byx = 1.5.

Now r^2 = byx * bxy = 1.5 * 0.5 = 0.75.

r = +sqrt(0.75) = 0.866 approx, positive because both coefficients are positive.

Answer: 2x - y = 5 is the line of x on y, 3x - 2y = 4 is the line of y on x, and r = 0.866.

TOPIC 9

9

Comparison of Correlation and Regression

Correlation measures strength and direction; regression gives the exact equation for prediction.

Correlation is symmetric (r for x,y equals r for y,x); regression is not (y on x differs from x on y).

Correlation needs both variables random; regression treats x as the fixed independent variable.

QUESTION BANK

30 **Distinguish between correlation and regression.**

IMPORTANT

Correlation measures the strength and direction of the relationship between two variables; regression gives the exact equation to predict one from the other.

Correlation is symmetric: the r between x and y is the same as between y and x. Regression is not symmetric: the line of y on x is different from the line of x on y.

Correlation lies between -1 and +1; regression coefficients can take any value and may exceed 1.

Correlation does not say which variable is cause and which is effect; regression treats x as the fixed independent variable and y as the dependent variable.

Correlation is not used for actual prediction, while regression is mainly used for estimation and forecasting.

Example: r = 0.9 only says marks and study hours move together strongly, while the regression equation lets us predict marks for 5 study hours.